**Reg. No.:**

**Name :**

## Mid-Term Examinations, April 2021

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|---|---|---|---|---|---|
| Programme | : | **B.Tech** | Semester | : | **Winter 2020-2021** |
| Course | : | **Differential And Difference Equations** | Code | : | **MAT2001** |
| Faculty | : | **Dr. A.Manickam** | Slot/Class No. | : | **E11+E12/0431** |
| Time | : | **1½ hours** | Max. Marks | : | **50** |

### Answer all the Questions

| Q. No. | Question Description | Marks | Module No. | RBT Level | CO |
|---|---|---|---|---|---|
| 1 | Diagonalise the matrix $A = \begin{pmatrix} 2 & 2 & -7 \\ 2 & 1 & 2 \\ 0 & 1 & -3 \end{pmatrix}$ by similarity transformation and hence find $A^4$ | **10** | | | |
| 2 | Find the Eigen values and eigen vectors of $A = \begin{bmatrix} 3 & 10 & 5 \\ -2 & -3 & -4 \\ 3 & 5 & 7 \end{bmatrix}$ | **10** | | | |
| 3 | Obtain the Fourier series expansion of $f(x) = x\sin x$ in $-\pi < x < \pi$. | **10** | | | |
| 4 | Find the first two harmonies of the Fourier series expansion for the data (see table below) | **10** | | | |
| 5 | Find the Fourier Transform of $f(x) = \begin{cases} 1-\lvert x\rvert; & \lvert x\rvert < 1 \\ 0 \quad ; & \lvert x\rvert > 1, \end{cases}$ Hence deduce that (i) $\int_0^\infty \left(\frac{sint}{t}\right)^2 dt = \frac{\pi}{2}$ (ii ) $\int_0^\infty \frac{sin^4 t}{t^4} \mathrm{dt} = \frac{\pi}{3}$ | **10** | | | |

Data for Question 4:

| $x$ | 0 | $\pi/3$ | $2\pi/3$ | $\pi$ | $4\pi/3$ | $5\pi/3$ | $2\pi$ |
|---|---|---|---|---|---|---|---|
| $f(x)$ | 1.0 | 1.4 | 1.9 | 1.7 | 1.5 | 1.2 | 1.0 |

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